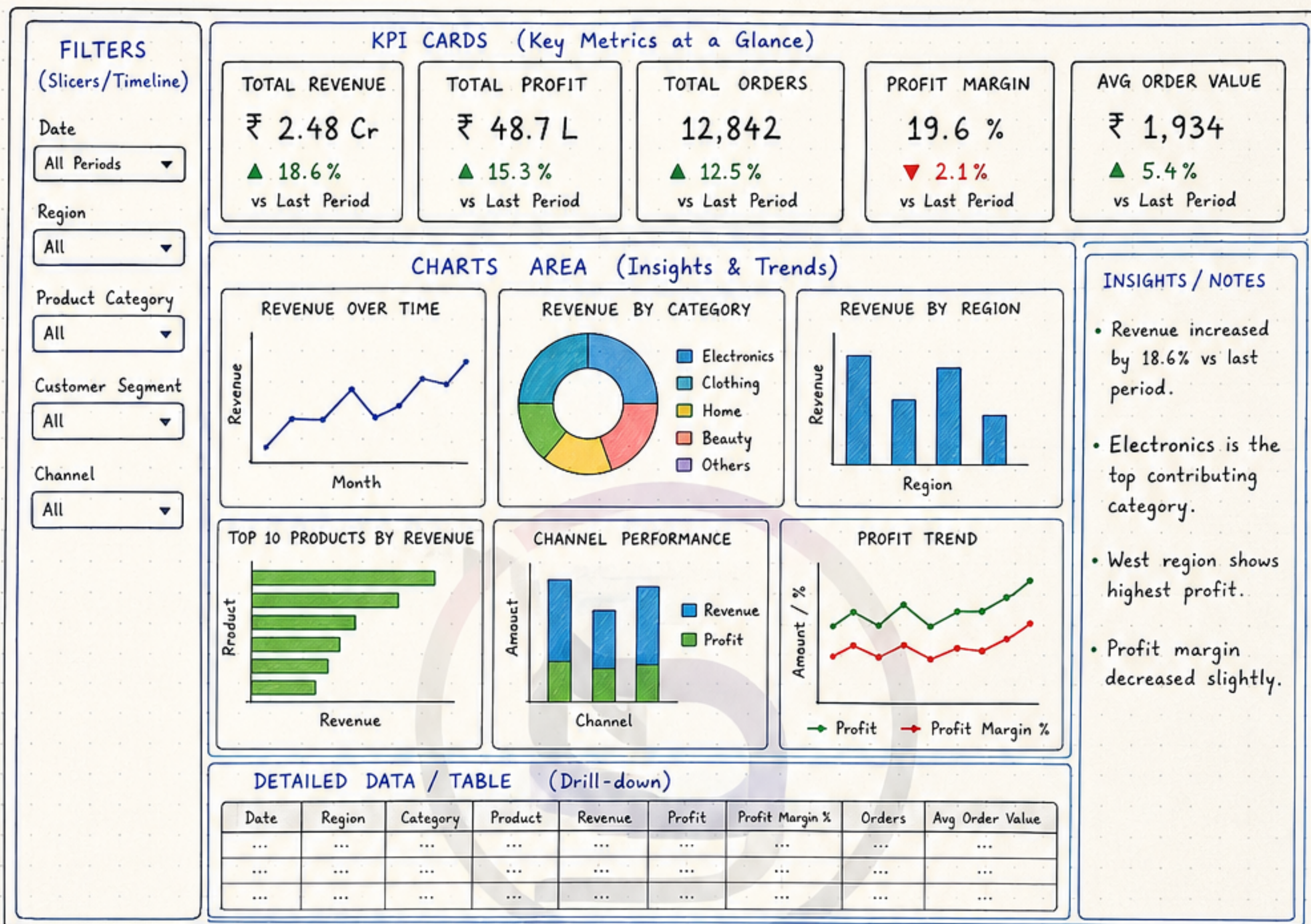


1. DASHBOARD LAYOUT FOR BUSINESS

DEFINITION: A Dashboard is a visual summary of important business metrics that helps stakeholders monitor performance and make data driven decisions.



2. CHARTS PLACEMENT STRATEGY

- Top Row → KPI Cards (High level summary)
- Left Side → Filters (User Control)
- Middle Area → Key Charts (Insights & Trends)
- Bottom → Detailed Table (Drill-down)
- Right Side → Insights / Commentary

WHY THIS LAYOUT?

- Follows Natural Reading Pattern (Left → Right, Top → Bottom)
- KPIs are the first things stakeholders look at.
- Charts provide visual insights.
- Filters allow interactivity.
- Table supports detailed analysis.
- Insights help storytelling.

3. KPI STRUCTURE

- KPI = Key Performance Indicator
- Should be simple, measurable, and actionable.
- Each KPI card should contain:
 - KPI Name
 - KPI Value
 - Trend Indicator (▲ / ▼)
 - % Change vs Last Period / Target

KPI CARD EXAMPLE

TOTAL REVENUE

₹ 2.48 Cr

▲ 18.6%
vs Last Period

Green = Good
Red = Bad

4. IMPORTANT KPIs FOR BUSINESS DASHBOARDS

SALES DASHBOARD

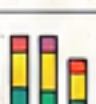
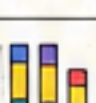
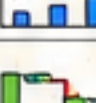
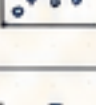
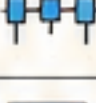
- Total Revenue
- Total Orders
- Average Order Value
- Total Customers
- Revenue Growth %
- Profit / Gross Profit
- Profit Margin %
- Top Products
- Top Categories
- Top Regions

FINANCE DASHBOARD

- Total Revenue
- Total Expenses
- Net Profit
- Profit Margin %
- Cash Flow
- Budget vs Actual
- Expense by Category
- Receivables
- Payables
- Working Capital

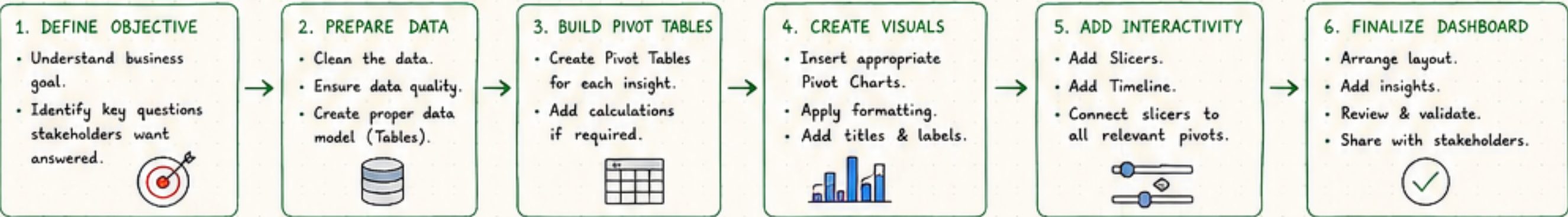
5. CHART SELECTION GUIDE (INTERVIEW FAVOURITE)

Choose the RIGHT chart based on the business question, data type & insight you want to communicate.

BUSINESS REQUIREMENT	BEST CHART	WHY THIS CHART?	EXAMPLE USE CASE
Compare Categories	 Column Chart	Easy comparison of values across categories.	Revenue by Region, Sales by Product
Ranking / Top N	 Bar Chart	Better for long labels, easy ranking.	Top 10 Products, Top Merchants
Trend Over Time	 Line Chart	Shows pattern, trend, seasonality clearly.	Monthly Revenue, Daily Orders
Part-to-Whole (Few Categories)	 Pie Chart	Shows contribution of each part to the total.	Sales by Channel, Market Share
Part-to-Whole (Many Categories)	 Doughnut Chart	Similar to pie but looks cleaner for dashboards.	Expense Share, Revenue Mix
Composition + Comparison	 Stacked Column	Shows total + contribution of each part.	Sales by Region by Category
% Composition Comparison	 100% Stacked Column	Compares percentage mix across categories.	Market Share by Region
Two Metrics with Different Scales	 Combo Chart	Combines two chart types with secondary axis.	Revenue & Profit %, Sales & Growth %
Increase / Decrease (Bridge Analysis)	 Waterfall Chart	Shows how a value changes from start to end.	Revenue to Net Profit, Budget vs Actual
Relationship Between Two Variables	 Scatter Chart	Shows correlation between numeric variables.	Discount vs Profit, Age vs Income
Distribution of Numeric Data	 Histogram	Shows frequency distribution of values.	Transaction Amount Distribution
Spread, Quartiles & Outliers	 Box & Whisker	Shows median, quartiles, spread & outliers.	Transaction Amount by Merchant Category
Funnel / Conversion Stages	 Funnel Chart	Shows drop-off at each stage.	Lead Conversion, Recruitment Funnel
Hierarchical Contribution	 Treemap	Shows hierarchy with rectangle size based on value.	Category → Subcategory Sales
Geographic Analysis	 Map Chart	Shows data on map by location.	Sales by State, Fraud by Country
Multi-metric Comparison	 Radar Chart	Compares multiple metrics across entities.	Employee Performance, Department Score

NOTE: Always choose the simplest chart that communicates the insight clearly.

8. DASHBOARD CREATION PROCESS



9. IMPORTANT TIP

A good dashboard is not about showing MORE data, it is about showing the RIGHT data that leads to BETTER decisions.

10. BUSINESS USES OF DASHBOARDS
- ★ Sales Performance Monitoring

★ Financial Reporting & Analysis

★ Operations & Process Monitoring

★ Marketing Campaign Analysis

★ Customer & Merchant Analysis

★ Executive Reporting

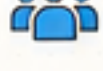
★ Fraud & Risk Monitoring

★ Project Tracking & KPI Management



11. KPI DEEP DIVE – WHAT KPIS SHOULD YOU INCLUDE?

Choose KPIs based on business goals & stakeholder needs.

BUSINESS FUNCTION	PRIMARY GOAL	IMPORTANT KPIS
Sales 	Increase Revenue Grow Customers	- Total Revenue - Total Orders - Average Order Value - Revenue Growth % - Top Products - Top Regions
Finance 	Improve Profitability Optimize Costs	- Total Profit - Profit Margin % - Expenses - Net Profit - Cash Flow - Budget vs Actual
Marketing 	Increase Leads Improve Conversion	- Total Leads - Conversion Rate % - Cost per Lead - Campaign ROI - Engagement Rate
Operations 	Improve Efficiency Reduce TAT & Errors	- Total Orders Processed - Avg Processing Time - Order Accuracy % - SLA Achievement % - Return Rate %
Customer Service 	Improve Customer Satisfaction & Retention	- Total Tickets - Resolution Rate % - Avg Resolution Time - Customer Satisfaction (CSAT) - Repeat Customer %
HR 	Improve Employee Performance & Retention	- Headcount - Attrition Rate % - Avg Performance Score - Attendance % - Training Hours
E-commerce 	Grow Sales & Improve Customer Experience	- Total Revenue - Add to Cart Rate % - Checkout Conversion % - Abandonment Rate % - Refund Rate %

NOTE: Keep KPI count limited (5 to 8) for clarity & focus.

14. COLOR & FORMATTING GUIDE

COLOR PSYCHOLOGY

- Blue – Trust, Professional, Corporate
- Green – Growth, Positive, Success
- Orange – Energy, Attention, Action
- Red – Alert, Warning, Important
- Gray – Neutral, Balanced

BEST PRACTICES

- Use max 5-6 colors in a dashboard.
- Use consistent colors for same metrics.
- Use light background, dark text.
- Highlight important numbers only.
- Avoid bright & clashing colors.
- Use conditional formatting wisely.

FORMATTING TIPS

- Use short titles & labels.
- Use consistent font size.
- Align charts & visuals properly.
- Use thousand separators (1,00,000).
- Show % with % sign, currency with ₹.
- Remove gridlines in charts (if not required).

12. FILTERS (SLICERS & TIMELINE) –BEST PRACTICES

DO'S

- Use Slicers for categories (Region, Product, Channel).
- Use Timeline for date fields (Year, Quarter, Month).
- Place filters on the left or top for easy access.
- Connect slicers to all relevant Pivot Tables.
- Use 'Clear Filter' option.
- Use consistent naming.
- Keep slicers simple – less is more.

DON'TS

- Don't overuse slicers.
- Don't connect slicers to unnecessary pivots.
- Don't use too many items in one slicer.
- Don't place slicers at the bottom.
- Don't use slicers that destroy comparison in charts.

WHEN NOT TO CONNECT SLICERS?

- If the slicer field is the same as the chart dimension (e.g., Chart shows Revenue by Region and slicer is Region) → It will filter chart to only one item and comparison will be lost.
- If slicer does not add any meaningful filtering.
- If it confuses users or makes the dashboard cluttered.

13. DASHBOARD INTERACTIVITY

DO'S

- Use slicers & timeline for dynamic filtering.
- Charts update automatically.
- Provide drill-down for detailed analysis.
- Use Pivot Charts for dynamic dashboards.
- Use tooltips for extra insights.
- Add buttons for navigation (if multi-page).

BENEFITS

- Better user experience
- Quick decision making
- Real-time insights
- Self-service reporting
- Improved engagement

15. COMMON DASHBOARD MISTAKES

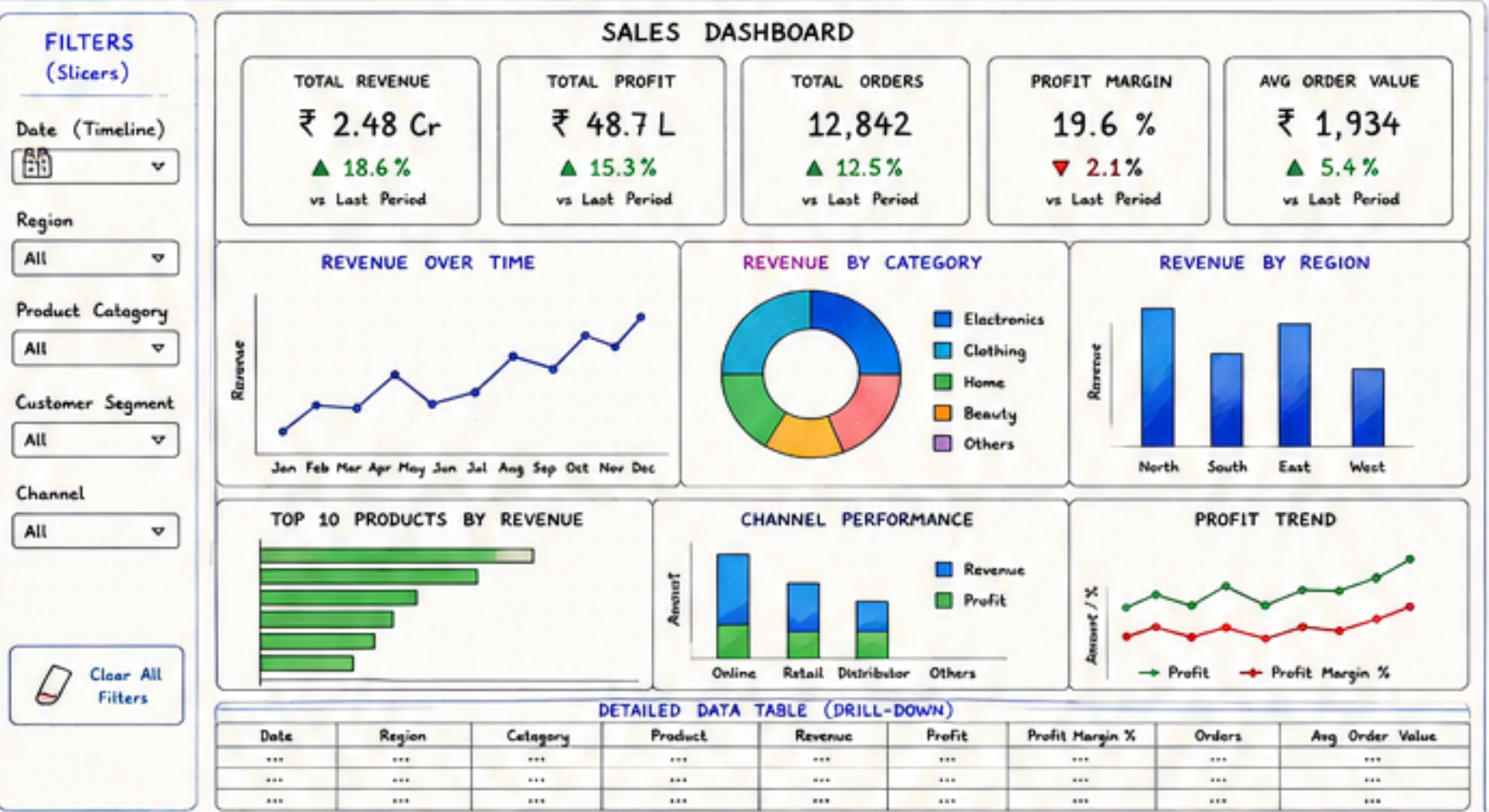
DO NOT DO

- Too many KPIs.
- Too many colors.
- Overcrowded charts.
- Wrong chart selection.
- Poor data quality.
- Missing titles & labels.
- No insights or commentary.
- Not mobile friendly.
- Ignoring stakeholder needs.

HOW TO AVOID?

- Keep it simple.
- Follow chart selection guide.
- Focus on key metrics.
- Validate data regularly.
- Design for the end user.
- Add meaningful insights.
- Review & improve regularly.

16. EXAMPLE – SALES DASHBOARD (WHAT IT SHOULD SHOW)



17. STAKEHOLDER INSIGHTS (WHAT THEY GET)

DO'S

- Quick performance overview in top KPI cards.
- Trends & patterns over time.
- Best & worst performing products/regions.
- Category & channel contribution.
- Ability to filter & drill-down.
- Actionable insights for better decisions.

BUSINESS IMPACT

- ★ Better Decisions
- ★ Higher Efficiency
- ★ Increase Revenue
- ★ Reduce Costs
- ★ Improve Customer Satisfaction
- ★ Data Driven Culture



Remember:

A dashboard is not about showing data, it's about driving decisions!

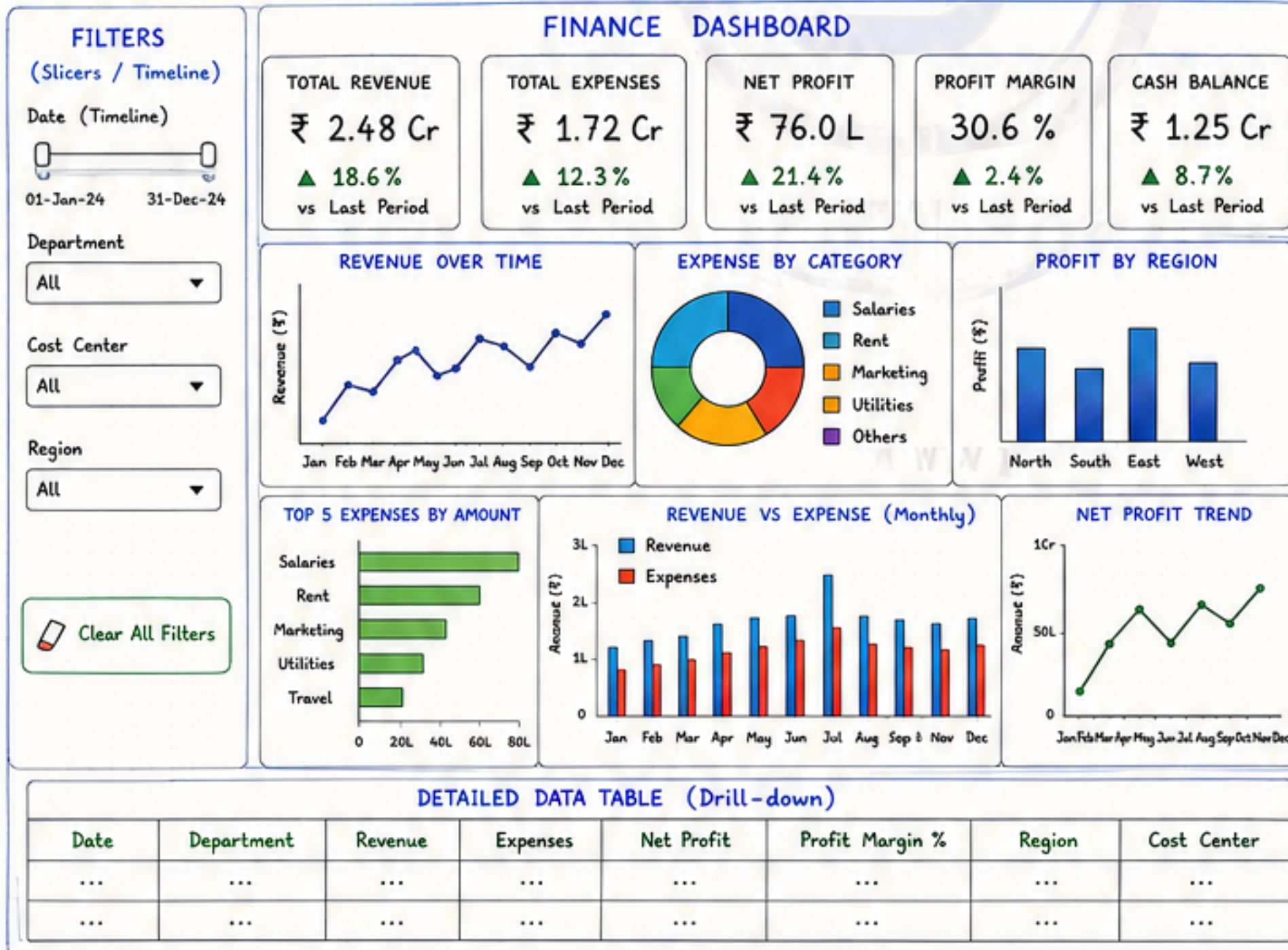
18. ADVANCED DASHBOARD FEATURES

- **Dynamic Titles** → Change based on slicer selection
(Use `=CELL("filename") & TEXT(TODAY(),"dd-mmm-yyyy")`)
- **Conditional Formatting** → Highlight important values automatically.
- **Drill-down** → Move from summary to detailed level.
- **Tooltips** → Provide extra information on hover.
- **KPI Trend Sparklines** → Show trends inside KPI cards.
- **Data Validation** → Ensure clean & consistent data.
- **Named Ranges** → Make formulas & pivots easy to manage.
- **Protect & Share** → Protect structure, allow user interaction.

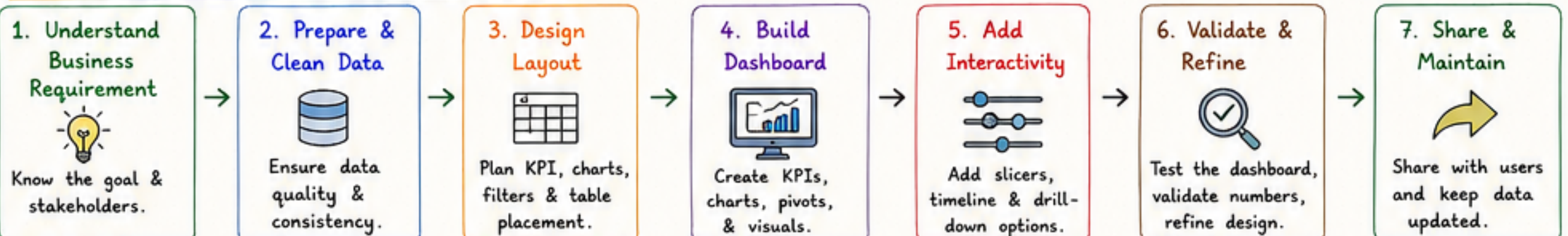
20. INTERVIEW QUESTIONS (DASHBOARD DEVELOPMENT)

- 1 What is the purpose of a dashboard?
- 2 Which KPIs would you include in a Sales dashboard and why?
- 3 How will you choose the right chart for a business question?
- 4 How do slicers and timeline improve dashboard interactivity?
- 5 What are the common mistakes in dashboard design?
- 6 How will you ensure data accuracy in a dashboard?
- 7 How would you make a dashboard user-friendly?
- 8 How do you handle large datasets in dashboards?

22. EXAMPLE - FINANCE DASHBOARD LAYOUT



25. DASHBOARD DEVELOPMENT ROADMAP



19. DASHBOARD CHECKLIST BEFORE PUBLISHING

- ✓ Is the purpose of the dashboard clear?
- ✓ Are all KPIs accurate and validated?
- ✓ Are charts answering the business questions?
- ✓ Are slicers and timeline working properly?
- ✓ Is the layout clean and easy to read?
- ✓ Are colors, fonts and labels consistent?
- ✓ Is the data up-to-date?
- ✓ Does the dashboard work on all screen sizes?
- ✓ Have you removed unnecessary clutter?
- ✓ Is the story clear from top to bottom?

21. DASHBOARD DEVELOPMENT PRINCIPLES

	Purpose First Every dashboard should answer a business question.
	Know Your Audience Design for stakeholders, not for yourself.
	Keep it Simple The best dashboard is easy to understand in 5 seconds.
	Focus on Insights Show WHAT happened and WHY it matters.
	Follow a Story Flow Guide users from overview → details → insights.
	Trust the Data Accuracy, consistency and timeliness are non-negotiable.

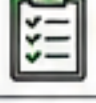
23. DASHBOARD TIPS

- ★ One dashboard = One purpose
- ★ Use white space effectively
- ★ Keep KPI cards at the top
- ★ Use consistent color theme
- ★ Avoid too many charts
- ★ Use icons to improve readability
- ★ Highlight exceptions
- ★ Test with real users
- ★ Keep refreshing data source
- ★ Automate where possible (Power Query / Power Pivot)

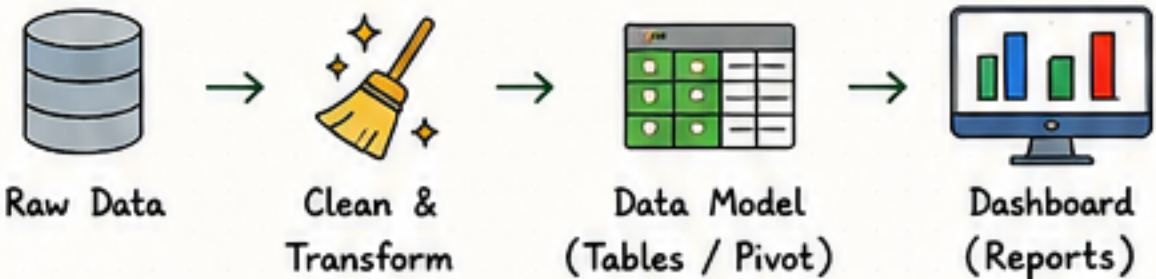
24. SUMMARY

A good dashboard is not about showing more data, it is about showing the **RIGHT** data in the **RIGHT** way to the **RIGHT** people at the **RIGHT** time to drive better business decisions.

26. DASHBOARD TYPES (WITH USE CASES)

DASHBOARD TYPE	DESCRIPTION	USE CASE / BUSINESS EXAMPLE
 Strategic Dashboard	High-level overview for executives.	CEO Dashboard – Revenue, Profit, Growth, Market Share.
 Operational Dashboard	Day-to-day operations tracking.	Operations Dashboard – Orders, SLA, Inventory, Fulfillment.
 Analytical Dashboard	Deep dive into data for analysis.	Sales Analysis – Trends, Category Performance, Customer Insights.
 Tactical Dashboard	Monitors short-term goals and targets.	Marketing Campaign Dashboard – Leads, Conversion, ROI.
 KPI Dashboard	Tracks key metrics at a glance.	KPI Dashboard – Revenue, Profit Margin, Orders, NPS, CSAT.
 Project Dashboard	Tracks project health, progress & issues.	Project Dashboard – Tasks, Timeline, Budget, Risks, Status.
 Customer Dashboard	Customer-centric performance view.	Customer Dashboard – Acquisition, Retention, LTV, Satisfaction.

27. DATA FLOW FOR DASHBOARD



BEST PRACTICES

- ✔ Use consistent data source for all visuals.
- ✔ Keep data updated & refreshed regularly.
- ✔ Avoid manual data entry wherever possible.
- ✔ Use Excel Tables for dynamic range.
- ✔ Use Power Query for cleaning & transformation.
- ✔ Use Data Model for large datasets & relationships.
- ✔ Test data accuracy before publishing.

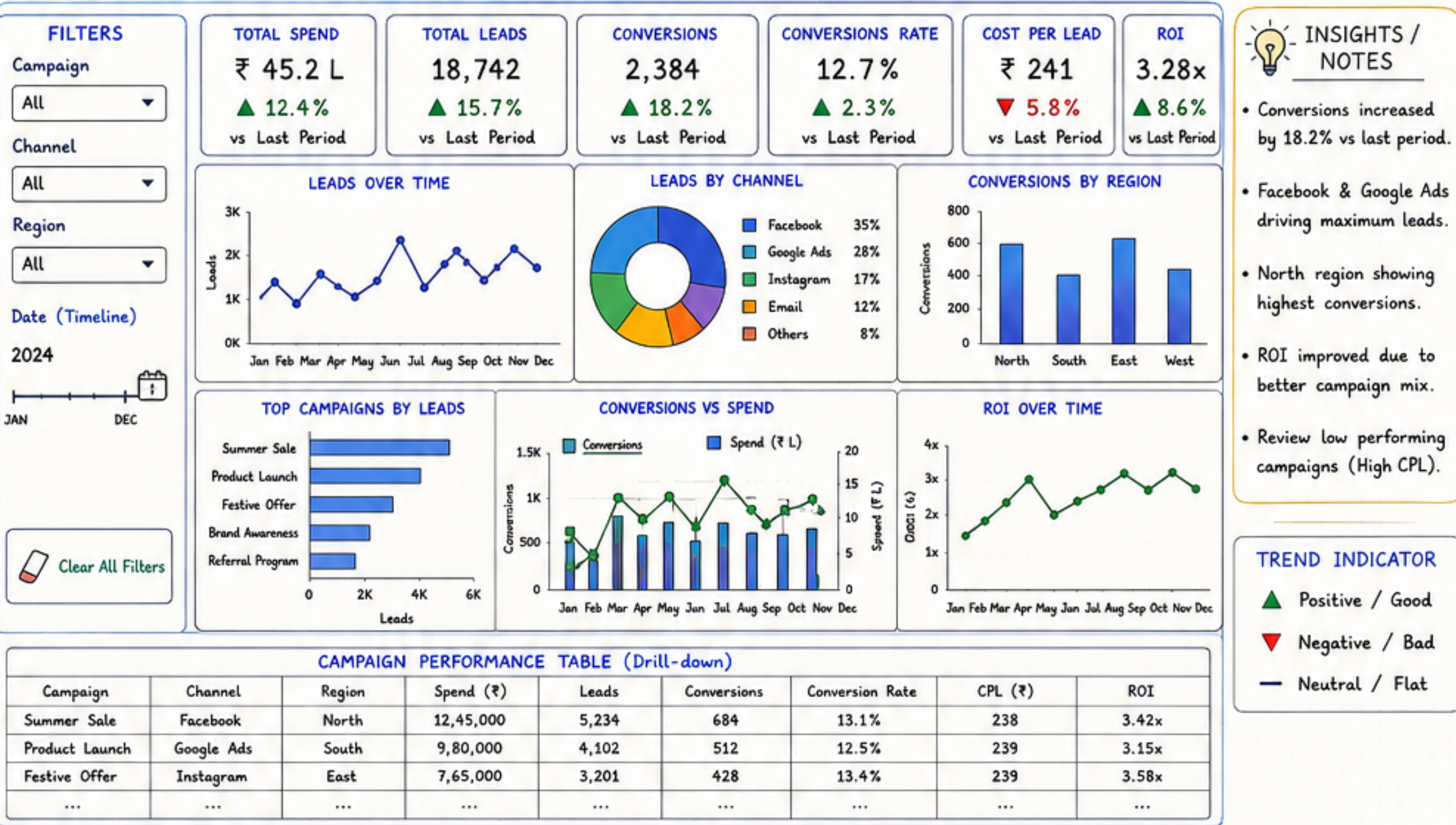
28. DASHBOARD PERFORMANCE TIPS

- ★ Avoid volatile functions (NOW(), TODAY(), RAND()).
- ★ Use Data Model instead of multiple large Pivot Tables.
- ★ Limit number of visuals on a single dashboard.
- ★ Use slicers wisely – fewer slicers, better performance.
- ★ Remove unnecessary calculations.
- ★ Use Power Query for heavy transformations.
- ★ Disable auto refresh while designing.
- ★ Optimize file size – remove unused data & formats.

29. INTERACTION ELEMENTS IN DASHBOARDS

	Slicers	Filter data interactively by categories (Region, Product, Channel, etc.)
	Timeline	Filter data by date (Year, Quarter, Month, Day).
	Buttons	Navigate between different dashboard pages or views.
	Dropdowns	Allow user selection from list.
	Drill-down	Move from summary → details.
	Tooltips	Show additional information on hover.

30. EXAMPLE – MARKETING CAMPAIGN DASHBOARD



31. KEY TAKEAWAYS

- ✔ A dashboard is a data story – keep it clear, simple & focused.
- ✔ Start with KPIs, then add charts that explain the KPIs.
- ✔ Use the right chart for the right question.
- ✔ Interactivity (Slicers, Timeline, Drill-down) makes dashboards powerful.
- ✔ Consistency in design builds trust and professionalism.
- ✔ Always validate data and test before sharing.
- ✔ A good dashboard saves time, improves decisions & drives results.

★ Remember:

Metrics show numbers,

Dashboards show the **story** behind the numbers!

32. ADVANCED CONCEPTS

- Dynamic Measures (DAX in Power Pivot / Data Model) for advanced KPIs.
- What-If Analysis** – allow users to test scenarios.
- Decomposition Tree** – drill into data hierarchies.
- Custom Sorting** – show data in meaningful order.
- Conditional Icons** – improve quick understanding.
- Bookmark Navigation** – switch views/pages instantly.
- Mobile Optimization** – create clean mobile-friendly layouts.
- Row Level Security (RLS)** – secure data for users.

WHY USE THESE?

- Provide deeper insights.
- Enable scenario planning.
- Improve user experience.
- Support decision making at multiple levels.
- Make dashboards scalable, secure & professional.



33. DASHBOARD DESIGN DO'S & DON'TS

DO'S

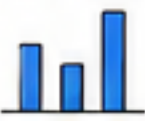
- Do keep it simple & focused.
- Do show only relevant KPIs.
- Do use consistent formatting.
- Do use meaningful titles.
- Do follow a logical flow.
- Do highlight key insights.
- Do use enough white space.
- Do test with real users.
- Do update data regularly.
- Do tell a clear data story.

DON'TS

- Don't add too many KPIs.
- Don't overcrowd with charts.
- Don't use 3D or fancy unnecessary visuals.
- Don't use inconsistent colors.
- Don't show unclear titles.
- Don't mix units in one chart.
- Don't ignore data quality.
- Don't make it slow with heavy data.
- Don't design for yourself – design for stakeholders.

34. DASHBOARD MATURITY LEVELS

1. BASIC REPORTING



- Static reports
- Manual updates
- Limited insights
- Descriptive analysis

FOCUS: What happened?

2. INTERACTIVE DASHBOARD



- Filters & slicers
- Dynamic visuals
- Drill-down capability
- Better user control

FOCUS: What is happening?

3. INSIGHTFUL DASHBOARD



- KPIs with targets
- Trend & variance analysis
- Root cause indicators
- Actionable insights

FOCUS: Why is it happening?

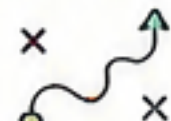
4. PREDICTIVE DASHBOARD



- Forecasts & predictions
- What-if scenarios
- Risk identification
- Future visibility

FOCUS: What will happen?

5. PRESCRIPTIVE DASHBOARD



- Recommendations
- Optimization
- Automated alerts
- Decision guidance

FOCUS: What should we do?

35. STAKEHOLDER PERSPECTIVE

STAKEHOLDER	WHAT THEY CARE ABOUT
CEO / Director	High-level KPIs, growth, profit, performance vs target, key risks.
CFO / Finance	Revenue, profit, cost, cash flow, budget vs actual, financial health.
Marketing Head	Leads, conversion, campaign ROI, customer acquisition cost.
Operations Head	Orders, SLA, inventory, process efficiency, errors, productivity.
Sales Manager	Sales, pipeline, targets, top products, lead status, win rate.
Customer Support	Tickets, resolution time, satisfaction, repeat issues, CSAT.

36. CHOOSE THE RIGHT KPIs – SMART

S	Specific	Clearly defined KPI that answers a business question.
M	Measurable	Quantifiable with numbers.
A	Achievable	Realistic and achievable.
R	Relevant	Relevant to business goals.
T	Time-bound	Has a specific time period.

EXAMPLE – BAD vs GOOD KPI

BAD KPI

Increase Sales



GOOD KPI

Increase Sales by 15% by Q2 2024

37. USING CONDITION FORMATTING IN DASHBOARDS

- Highlight Top / Bottom Performers (e.g., Top 5 Products)
- Traffic Lights for KPI Status (Green – Good, Yellow – Watch, Red – Bad)
- Data Bars for quick comparison
 – 65%
- Color Scales for intensity (Low → High)
- Icon Sets for trends

Result: Faster insights, better decisions.

38. RESPONSIVE DASHBOARD CHECKLIST

- Works on all screen sizes (Desktop / Tablet / Mobile)
- Clear fonts & readable text
- Slicers easy to use on mobile
- No horizontal scroll
- Important KPIs visible without scrolling
- Buttons & navigation optimized
- Performance tested with large data



39. TOOLS & TECHNOLOGIES (BEYOND EXCEL)

Power BI
Advanced dashboards, real-time data, sharing & collaboration.

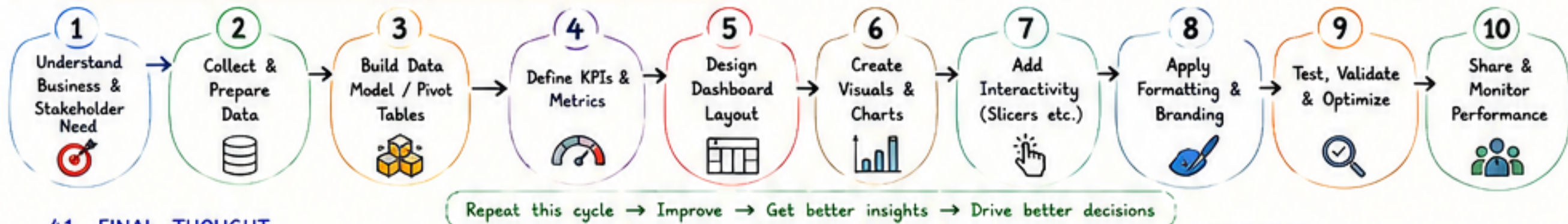
Tableau
Powerful visualization, analytics & discovery.

Looker Studio (Google Data Studio)
Free tool for reports, dashboards & data storytelling.

SQL
Extract, transform & load data for dashboards.

Python
Data analysis, automation & advanced visuals.

40. END-TO-END DASHBOARD DEVELOPMENT SUMMARY



41. FINAL THOUGHT



A dashboard is more than charts and numbers. It is a **story** that turns data into **insights** and insights into **actions**. Focus on **clarity**, **simplicity** and **value** for stakeholders.

QUICK RECALL

- ★ Right Data
- ★ Right KPIs
- ★ Right Charts
- ★ Right Layout
- ★ Right Insights
- ★ Right Decisions

REMEMBER



You don't build dashboards for data. You build dashboards for **people** to **make better decisions**.